CPSC 231 Fall 2026 Assignment 1 — Chart Drawing

It may become necessary to amend this specification at some point. You will be notified via D2L if that is the case.

Due Date: Thursday, Oct 1, 2026 @ 23:59
If the final submission of your assignment is **at least 24 hours before the due date**, you will receive **5 reward points**.

Late submission: -10 marks/day (minimum unit is *day*); no assignments will be accepted over two days late. You may redeem 10 reward points to remove 1 day late penalty (maximum 2 days for 20 points).

Overview

This assignment helps you learn **basic programming concepts,** including reading inputs, writing outputs, assigning values to variables, and using an external library. Use the *SimpleGraphics* library (you can download the library from D2L) to draw a chart. You can work on this assignment **individually** or in **pairs** — you can choose your partner, but your partner **must register in and attend the same tutorial** as you.

Policy

Violation of any of the following points is considered academic misconduct.

- You (and your partner) must not submit or look at solutions to the assignment problems that are not your own.
- You (and your partner) must not share your solution code with other students, nor ask others to share their solutions with you (and your partner) .
- You (and your partner) must indicate on your submission any assistance received.

Submission

Submit your Python program — **.py file** — to the **dropbox on D2L**. **Email submissions will NOT be accepted**. You should **only submit one file** that is the Python program and should not include other files. If you are working with a partner, only one of you should do the submission and make sure you **add both of your names and student ID numbers** in the comment box when you submit your assignment.

If you have any questions regarding the specifications for this assignment or how to submit it, please ask your TA because **it is your TA who will mark your assignment**. Your TA may also have specific requirements that you should follow.

Deliverables [Total 60 marks]

A. Program [30 marks]

Write a Python 3 program using the *SimpleGraphics* library to draw a **histogram** or a **bar chart** based on users' specifications. A histogram does not have space between columns, but a bar chart does. You can decide which chart to draw, but it must meet the following requirements:

- have three columns
    - heights and colours of the columns are decided by the user
    - must be roughly evenly distributed and centered on the x-axis
- have a title
    - centred at the top of the chart, regardless of length
    - the title is decided by the user
- have an x-axis and a y-axis
    - width of the chart is 400 pixels
    - height of the chart (excluding the title) is 300 pixels
- drawn at a location specified by the user

Note: For reference, you can find some sample inputs and charts at the end of this document.

Your program must ask the user to enter the following **in the following order**:

1. the title of the chart
2. values for the three columns
    - you can assume the values entered are integers
    - e.g., if the user enters 250, the column will be 250 pixels high
3. colours for the three columns
    - colours entered are one of the colours from pages 7 and 8 of the tutorial.pdf (available from D2L), e.g., *green* and *steel blue*, etc.
4. location of the chart
    - indicated by the x and y coordinates of the bottom-left corner of the chart. Note that the top-left of the whole canvas is (0,0). Please refer to Section 2.1 in tutorial.pdf for more information.

Note: the users will enter their input in the terminal, not in the graphics window.

For full marks, your program must:

- have information about you (and your partner) in the form of comments at the top of the program, including the rough proportion of the work distributed between you and your partner (if you worked with a partner). For example,
    - # Name (ID): Arthur Fleck (1357911) and Harley Quinn (246810)
    - # Tutorial: 01
    - # Arthur Fleck did about 40%. Harley Quinn did about 60%.
- **not** change the default window size created by the *SimpleGraphics* library.
- be properly commented. This means
    - making comments **only** where the code needs further explanation.
    - comment blocks of code to tell us what the blocks do.
    - do not comment every single line of code.
    - comments are formatted and organized in a way that makes them easy to follow.
    - good example:
        - # Draw a rectangle
        - setOutline("red")
        - setFill("light cyan")
        - rect(300, 100, 200, 100)
    - bad example:

        - setOutline("red") # draw red outline
        - setFill("light cyan") # fill with light cyan
        - rect(300, 100, 200, 100) # draw a rectangle
- have proper code style. This means your code
    - is organized and well thought out,
    - uses meaningful variable names, and
    - makes proper use of constants.
- have proper prompts in the correct order to guide users for inputs.
- apply the appropriate castings to all inputs, with no unnecessary casting.

You can decide:

- to draw a histogram or a bar chart
- if it is a bar chart, the spacing between the columns
- the width of the columns, but all columns **must** have the same width
- the font for the title
- whether to have outlines for the columns and the colours for the outlines
- whether to have labels for the axes

B. Demo [30 marks]

Done during your tutorial after the submission deadline.

The specific day(s) dedicated to demo will take place starting on the **first tutorial following submission** and continuing until every student has demoed, unless otherwise communicated by your TA. Your TA will be in contact with you about how to schedule your demo during these times - if you have any questions about this, please direct any and all questions to your assigned TAs.

During your assigned demo slot, your TA will ask you to enter specific values as your program inputs, and you will receive marks based on whether or not your output follows expectations. Your TA may also ask you questions about your code.

C. Partnering [up to 5 reward points]

If you choose to work in a pair, you will be eligible for up to a maximum of 5 reward points (RPs). These points will be distributed according to the rough proportion of the work distributed between you and your partner, as described in the deliverables above.

If students ***A*** and ***B*** worked together such that ***A*** did ***A%*** of the work while ***B*** did ***B%*** of the work, the RPs would be calculated as follows:

$$\text{A's reward points} = \lfloor \frac{A\%}{\max(A\%, B\%)} * 5 \rfloor$$

$$\text{B's reward points} = \lfloor \frac{B\%}{\max(A\%, B\%)} * 5 \rfloor$$

For example,

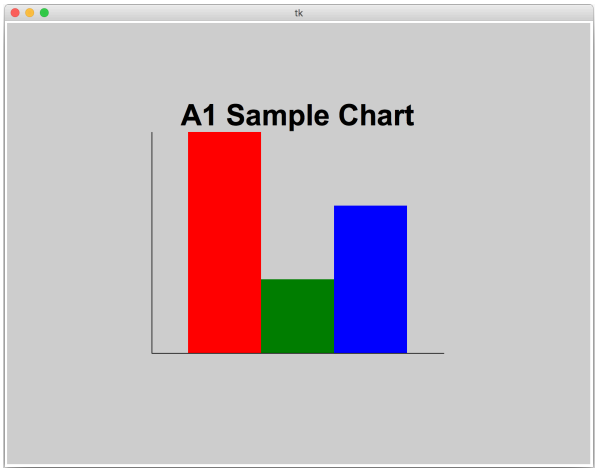
- Clark did about 40%; Bruce did about 60%.
    - Clark would receive 3 RPs while Bruce would receive the maximum 5.
- Both Tony and Steve did 50% each.
    - Both Tony and Steve would each be awarded 5 RPs.
- Sergio Marquina did 90%; Raquel Murillo did 10%.
    - Sergio Marquina would get the maximum 5 RPs while Raquel Murillo would get 0.

This breakdown **must** be included in the comment at the top of your program in order for the RPs to be distributed; **if the appropriate percentages are not listed, no RPs can be given to either student**.

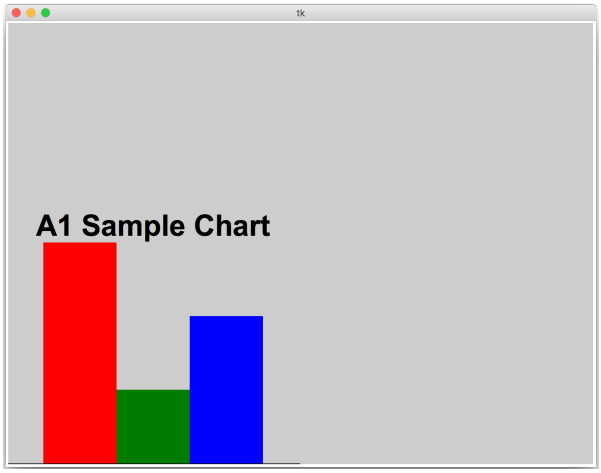
See sample inputs and charts on the next page.

Sample Inputs and Charts

```
Enter the title: A1 Sample Chart
Enter the value of bar 1:  300
Enter the value of bar 2:  100
Enter the value of bar 3:  200
Enter the colour of bar 1: red
Enter the colour of bar 2: green
Enter the colour of bar 3: blue
Enter the x-coordinate: 200
Enter the y-coordinate: 450
```




```
Enter the title: A1 Sample Chart
Enter the value of bar 1:  300
Enter the value of bar 2:  100
Enter the value of bar 3:  200
Enter the colour of bar 1: red
Enter the colour of bar 2: green
Enter the colour of bar 3: blue
Enter the x-coordinate: 0
Enter the y-coordinate: 600
```




```
Enter the title: A1 Sample Chart
Enter the value of bar 1:  300
Enter the value of bar 2:  100
Enter the value of bar 3:  200
Enter the colour of bar 1: red
Enter the colour of bar 2: green
Enter the colour of bar 3: blue
Enter the x-coordinate: 600
Enter the y-coordinate: 300
```

